EXPLAINABLE REVIEWED-ITEM RETRIEVAL, WITH A THAI TASTE CASE STUDY

Progress Report

(Proposal title: *Explainable Reviewed-Item Retrieval: Searching Restaurants by Taste Preference*)

by

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CHAPTER 1

RESPONSE TO PROPOSAL FEEDBACK

1.1 Feedback Summary Page

Table 1.1 answers every comment received on the proposal. Each row states what we understood the comment to mean, what we did, and where the evidence is. Detailed answers follow in Sections 1.2–1.6.

Table 1.1: Feedback summary: one row per comment from the proposal presentation.

Comment	What it means for us	Action taken / revision	Status
Strengths (problem formulation, RIRD labels, EDA and baselines, honest English/ Thai split)	Keep what worked.	Kept the reviewed-item retrieval formulation, RIRD as the only restaurant-level benchmark, and the rule that every claim is tied to a measurement.	Kept
F1. Application claims do not match the evaluation; RIRD is not Thai taste; Wongnai cannot support restaurant-level evaluation	Our "Thai food recommender" claim was unmeasured.	(1) Title and claims narrowed to what we measure. (2) Building a Thai taste test set: 24 Thai queries written and frozen by a native speaker before any results were seen; BM25 and dense sentence retrieval running on Wongnai; corpus fixed at 10,000 reviews; blind labelling sheet, 0/1/2 guidelines and scoring code ready. (3) Thai restaurant ranking declared future work (no restaurant ids exist).	Partly done; labelling next

Comment	What it means for us	Action taken / revision	Status
F2a. BM25 and dense score different units	The proposal compared one concatenated document (BM25) with mean-pooled review vectors (dense).	Revised set-up: both models score the same unit (review) with the same five aggregations, 95% bootstrap CIs and Holm-corrected paired tests. Dense beats BM25 under every aggregation (all significant).	Done (review level)
F2b. Fusion can be biased by review volume	Max and top-k get more "chances" for restaurants with more reviews (406–2,281).	Volume-control experiment: every restaurant capped at 400 reviews. Bias exists in the scores (per-query r up to 0.36) but removing it changes nDCG@10 by at most 0.011 and does not change any conclusion.	Done (review level)
F3. No evidence that explanations are faithful or complete	A relevant-looking sentence is not proof that it caused the ranking, nor that it supports the query.	Separated the two questions: *faithfulness* (deletion test) and *support* (blind human 0/1/2 rating with kappa). The evidence shown for every ranking is now saved, and a first inspection already shows failure cases (Section 5.4).	Designed; evaluation next
F4. Check more literature (three papers)	Position against multi-aspect RIR, explanation criteria and the review-based survey.	All three added to related work, plus the expert-finding literature on volume bias and the faithfulness literature behind both explanation tests (Chapter 3). Korikov et al.'s LLM query decomposition is cited, not reimplemented (reason in Section 1.5).	Done
Final question. Do retrieved sentences genuinely support the rankings, and does the approach work on real Thai reviews with independent taste labels?	The project's final research question.	Adopted as RQ2 and RQ3 (Section 2.2). First half: deletion test and human rating. Second half: Thai test set, answered at sentence level.	Plan fixed; answer in final report

Comment	What it means for us	Action taken / revision	Status
Earlier advice ("basics over complexity"; circular flavour features)	Do not spread effort over too many add-ons.	Dropped the flavour-feature layer, Thailand Foods gazetteer, M-ABSA, rating-weighted fusion and the three-encoder grid; effort moved to fairness and explanation evaluation (Section 1.7).	Done

1.2 F1: Application Claims versus Evaluation

Understanding. The proposal promised a Thai taste recommender, but every number came from RIRD, which contains English Yelp reviews of Toronto restaurants and broad preference queries such as "I am on a budget". Nothing measured whether the system understands Thai taste words (เปรี้ยว, เผ็ด, ไม่หวานมาก). Wongnai, our only Thai corpus, has review text and stars but no restaurant identifier, so it cannot be used to rank restaurants at all.

Actions.

1. **Narrowed the claim.** The title is now *Explainable reviewed-item retrieval, with a Thai taste case study.* Restaurant ranking and explanations are claimed only for English (RIRD); Thai is claimed only at the sentence-retrieval step.
2. **Built a way to measure Thai.** A 24-query Thai taste test set over real Wongnai review sentences (Section 4.3): 11 single-taste, 8 combination and 5 negation queries. They were drafted, then reviewed and frozen by a native Thai speaker *before* any retrieval output was seen, so the queries cannot have been tuned to make one system look good.
3. **Stated the limit.** Thai restaurant ranking needs Thai reviews grouped by restaurant, which no public dataset provides; it is future work.

Why not translate RIRD into Thai? Translated reviews would test translation output, not how Thai diners actually write about food (แซ่บ, เผ็ดดดด, code-mixed English), and the translation quality would become an uncontrolled variable.

1.3 F2: Units and Review-Volume Bias

Understanding. The comment has two parts. (a) In the proposal, BM25 scored one long concatenated document per restaurant while dense retrieval scored individual reviews and averaged their vectors, so a difference between them could come from the unit rather than the model. (b) Late fusion is not neutral to the number of reviews: if a restaurant has n review scores, the expected maximum (and the mean of the top k) grows with n even when every review is equally relevant. For roughly Gaussian scores the expected maximum grows like $\sigma\sqrt{2\ln n}$, so a restaurant with 2,281 reviews has more chances to produce one lucky high score than one with 406. A plain mean does not have this problem. The same effect is known in expert finding, where "candidate length normalisation" is used to stop prolific candidates from winning by volume (Macdonald & Ounis, 2008).

Actions. (a) The revised set-up makes both models score the same unit with the same aggregation function (Section 5.2). (b) A volume-control experiment caps every restaurant at the same 400 randomly chosen reviews and re-aggregates the same unit scores, so only the volume changes (Section 5.3).

Answer. The volume bias is real in the scores but harmless to the ranking quality on RIRD. Restaurant scores under max and top-k correlate with review count within a query (Pearson up to 0.36) and this drops to about 0 after capping, yet nDCG@10 changes by at most 0.011 and the model ranking does not change. Part of the correlation is legitimate: restaurants with more reviews are also relevant to more RIRD queries (Spearman 0.30 across the 50 restaurants).

1.4 F3: Evidence for Explainability

Understanding. "Explanation quality" contains two different questions that the proposal mixed together (Jacovi & Goldberg, 2020):

- **Faithfulness:** do the shown sentences actually *cause* the restaurant's position, or would the ranking be the same without them?
- **Support (plausibility):** would a person reading the sentence agree that it shows the restaurant matches the query?

A sentence can pass one test and fail the other. Our first inspection shows exactly this: for the query "I am on a budget", the top-ranked restaurant's strongest evidence is *"It's not bad but kinda pricey."* It is the review closest to the query (it is about price),

so it faithfully reflects why the model ranked the restaurant first, but it contradicts the query, and the annotators marked that restaurant not relevant.

We also realised that the aggregation function decides whether a faithful extractive explanation is possible at all. Under max or top-k mean, the shown top units are exactly the inputs of the score. Under a plain mean, the score depends on all 400 or more reviews equally, so showing the top three is not faithful by construction. This is an argument for top-k mean beyond its accuracy.

Actions. Two tests, designed and ready to run on the saved evidence (Section 4.2): a deletion test adapted from ERASER's comprehensiveness (DeYoung et al., 2020); and a blind human 0/1/2 support rating by two annotators with Cohen's kappa, using the factuality-style criteria of Zhang et al. (2025). We only *select* existing review sentences; we do not generate explanations.

1.5 F4: Literature

All three suggested papers are now in Chapter 3. Korikov et al. (2024) show that top-K late fusion fails when a query has several aspects spread unevenly across reviews, and fix it with LLM query decomposition and per-aspect fusion. We cite it as the closest work on fusion behaviour but do not reimplement it: our question is whether fusion is *fair* with respect to volume and unit, not how to decompose queries; LLM decomposition would add cost and a non-reproducible component; and it has not been shown to work for Thai. It does predict that our multi-taste Thai queries (e.g. อะไรแซ่บๆ เปรี้ยวเผ็ดเค็ม) will be the hardest, which we will report as a separate slice. Zhang et al. (2025) supply criteria for judging explanations, which shaped our support-rating guidelines. Hasan et al. (2025) let us position the project: we are a query-driven, content-based use of reviews, not a rating-prediction recommender.

1.6 The Final Question

> *Do retrieved review sentences genuinely support the system's restaurant rankings, and can the same approach work on real Thai restaurant reviews with independently annotated taste preferences?*

We adopted this as the project's final research question. Its first half is answered by the deletion test and the human support rating on RIRD. Its second half is answered by the Thai test set, at the *sentence* level only, and we will say so explicitly. Current

status: the method runs end-to-end on real Thai reviews and the labelling protocol is ready; neither half has a measured answer yet.

1.7 Suggestions Not Taken and Items Dropped

Table 1.2: Items removed from the proposal scope, with reasons.

Item	**Reason**
LLM query decomposition (Korikov et al., 2024)	Different research question; cost and reproducibility; untested for Thai. Cited, not reimplemented.
Rating-weighted fusion	Mixes overall satisfaction (stars) into a relevance score and adds a second popularity-like bias, the opposite of what F2 asks for.
Three-encoder grid	One multilingual encoder is shared by the English and Thai tracks so they stay comparable; the time goes to F2 and F3 instead.
Flavour-feature layer, Thailand Foods, M-ABSA	Risk of circular validation; M-ABSA has no taste-intensity aspects. Replaced by direct human labels on Thai sentences.
Fine-tuning the encoder	Out of scope; all models are used off the shelf.

CHAPTER 2

PROJECT OVERVIEW

2.1 Problem

Given a free-text preference query q, rank restaurants and show the review sentences that justify each position. Each restaurant r_i has reviews, split into units $u_{i,j}$ (a review or a sentence). A retrieval model scores every unit, $s(q, u_{i,j})$, and an aggregation function f turns the unit scores into a restaurant score

$$\text{Score}(q, r_i) = f\big(\{\, s(q, u_{i,j}) \,\}_j\big), \qquad f \in \{\text{max},\ \text{mean},\ \text{top-}k\ \text{mean}\}, \tag{2.1}$$

and the highest-scoring units of each ranked restaurant are shown as its explanation.

2.2 Research Questions

RQ1 (fair fusion). When BM25 and dense retrieval score the same unit with the same aggregation, which is better, and does the answer survive a control for review volume?

RQ2 (explanations). Are the shown review sentences faithful to the ranking (deletion test) and do people judge them as supporting the query (human rating)?

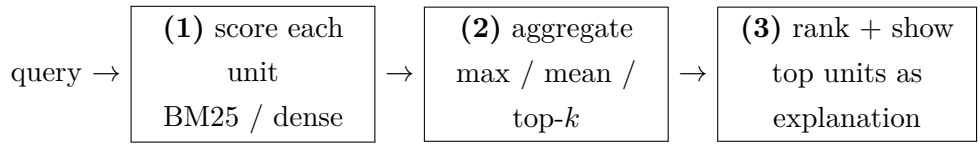
RQ3 (Thai). Does the sentence-retrieval step work on real Thai restaurant reviews, judged against independent native-speaker taste labels?

2.3 Datasets in Use

Table 2.1: Datasets and their role after the revision.

Dataset	Language	Structure	Role
RIRD (Abdollah Pour et al., 2023)	English	29,168 reviews, 50 restaurants (406–2,281 reviews each); 100 queries × 50 restaurants, 5 annotators	RQ1, RQ2: restaurant ranking and explanation evaluation
Wongnai (Highground-bkk, n.d.)	Thai	40,000 train reviews, text and 0–4 stars only; no restaurant ids	RQ3: sentence-level test set with our own queries and labels

2.4 Pipeline and Tracks



Track A (RQ1): stages 1–2 on RIRD
Track B (RQ2): stage 3 on RIRD
Track C (RQ3): stage 1 on Wongnai (Thai)

Figure 2.1: Pipeline and the stage each track evaluates.

The dense model is `paraphrase-multilingual-mpnet-base-v2` (Reimers & Gurevych, 2020), used for both languages; BM25 is `BM25Okapi` (Robertson & Zaragoza, 2009), with English lower-case tokens and Thai `newmm` word segmentation (Phatthiyaphaibun et al., 2023).

CHAPTER 3

RELATED WORK (UPDATED)

3.1 Reviewed-Item Retrieval and Multi-Aspect Fusion

Abdollah Pour et al. (2023) define reviewed-item retrieval, release RIRD, and compare early fusion (average the reviews, then score) with late fusion (score each review, then aggregate the top K); their best fine-tuned late-fusion model reaches MAP 0.626, and they also report that off-the-shelf dense retrieval beats BM25. Our dense versus BM25 result is therefore a *replication* with a multilingual encoder, not a new finding. Korikov et al. (2024) show that top-K late fusion breaks when a query's aspects are unevenly distributed across an item's reviews and propose LLM query decomposition with per-aspect fusion. Both papers score whole reviews; neither tests a volume control or a sentence unit, which is where our study adds evidence.

3.2 Aggregation and Volume Bias

Aggregating document scores into a score for an entity is an old problem in expert finding and blog distillation. Balog et al. (2006) compare one profile per candidate (like our proposal's concatenated document) with scoring documents and then aggregating (like our revised set-up), and find the second better. Macdonald & Ounis (2008) frame aggregation as voting (CombMAX, CombSUM and others) and show that count-based voting rewards prolific candidates, which motivates their normalisation. Our max and top-k mean are such voting functions, and our volume control is the reviewed-item version of this normalisation question. Finer retrieval units tend to help off-the-shelf dense retrievers (Chen et al., 2024), which motivates the sentence-level rerun.

3.3 Explanations and How to Evaluate Them

Selecting existing review sentences as explanations is an established framing (Li et al., 2021). Jacovi & Goldberg (2020) separate faithfulness (does the explanation

reflect the model's reasoning) from plausibility (is it convincing to people). ERASER (DeYoung et al., 2020) measures faithfulness by deleting the rationale and observing the change in the model's output (comprehensiveness); our deletion test applies it to a restaurant ranking. Zhang et al. (2025) argue that explanations should be judged from the user's side and define criteria such as factuality and personalisation; we borrow the factuality criterion ("is this sentence actually evidence for the query?") for our support rating. Hasan et al. (2025) survey review-based recommender systems and list explainability and evaluation as open challenges; most surveyed systems use reviews to predict ratings, whereas we use them to answer a free-text query.

3.4 Thai Retrieval

Thai retrieval benchmarks mostly cover Wikipedia, QA or legal text. On the Thai part of MIRACL (Zhang et al., 2023), BM25 (nDCG@10 0.484) beats an older zero-shot dense retriever (mDPR, 0.358), so we do *not* assume dense wins in Thai. Negation is hard for neural retrievers (Weller et al., 2024); our Thai set therefore has negation queries reported as their own slice. We found no Thai test set for taste-preference retrieval over restaurant reviews.

CHAPTER 4

UPDATED METHODOLOGY

4.1 Track A: Fair Fusion on RIRD

- **Shared unit.** One BM25 index over all 29,168 reviews (not one document per restaurant), so IDF separates common from rare words; dense embeds every review (max 256 tokens). Both produce a query × unit score matrix.
- **Shared aggregation.** The same function turns either matrix into restaurant scores: max, mean, and top-k mean for $k \in \{3, 5, 10\}$.
- **Metrics.** nDCG@10 (graded, annotator vote sum 0–5; Järvelin & Kekäläinen 2002), MAP, MRR, P@k and Recall@k (binary aggregated label). Positives are sparse (27%), so only rank-aware metrics are used.
- **Uncertainty.** 95% bootstrap CI over the 100 queries (10,000 resamples); paired sign-flip randomization test per comparison, Holm-corrected within each family (Smucker et al., 2007).
- **Floors.** Random order, review count, and leave-one-query-out popularity (the number of *other* queries a restaurant is relevant to). A model that does not beat popularity is mostly ranking restaurants that are relevant to everything.
- **Volume control.** Sample 400 reviews per restaurant (fixed seed; the smallest restaurant has 406), re-aggregate the *same* unit scores, and compare metrics and the per-query correlation between restaurant score and review count.
- **Saved evidence.** For every query, model and aggregation we store the full restaurant ranking and the top-10 evidence units with their text (50,000 ranking rows and 500,000 evidence rows), so the explanation tests use exactly what the ranker produced.
- **Next:** the identical pipeline at sentence level (Section 6).

4.2 Track B: Explanation Evaluation

Faithfulness: deletion test (automatic). For each query and each of the top-3 restaurants, delete the $m \in \{1, 3, 5, 10\}$ units shown as its explanation, recompute its score with the same aggregation, and record its new rank. Compare with deleting m random units of the same restaurant (averaged over 20 draws). A faithful explanation should cause a clearly larger rank drop than random deletion. We also report the change in overall nDCG@10 when the explanation units are removed from every restaurant. Under top-k mean, deleting the k shown units must change the score, so the informative quantity is *how far* the restaurant falls: if the next-best units are almost as good, the explanation is faithful but not unique, i.e. the evidence is redundant rather than decisive.

Support: human rating. 30 queries × top-3 restaurants × top-3 sentences (270 items), shown without system information. Two annotators label each item 0 (irrelevant or contradicts), 1 (partly supports) or 2 (clearly supports why this restaurant matches the query). We report Cohen's kappa (Cohen, 1960) and the share of items rated supporting, and break failures down by type (topic match with wrong polarity, negation ignored, wrong aspect).

4.3 Track C: Thai Taste Test Set

1. **Queries (done).** 24 Thai queries: 11 single taste (sour, spicy, sweet, rich, oily, cheap, salty, herbal, mild), 8 combinations (e.g. sour and spicy tom yum) and 5 negations (e.g. desserts that are not too sweet). Written in natural Thai, reviewed and frozen by a native speaker on 2026-10-17 before seeing any output.

2. **Sentences (done).** PyThaiNLP `crfcut` sentence splitting; fragments under 15 characters dropped. Spot-checked by a native speaker (Section 5.5).

3. **Corpus size (decided).** 10,000 Wongnai reviews (about 95,000 sentences), up from the 2,000-review pilot. Rare tastes and negations need enough truly relevant sentences in the corpus; otherwise both systems look bad for lack of matches, not lack of skill. The size is fixed *before* labelling because labels are tied to the corpus.

4. **Pooling.** Per query, the union of the BM25 top-10 and dense top-10 sentences, shuffled, with the source system hidden. With so little overlap between systems this gives about 18–20 sentences per query.

5. **Labelling.** 0/1/2 labels by a native speaker following written guidelines with decided rules for every query (e.g. for "not too sweet", หวานเกินไป = 0, ไม่หวานมาก = 2; for a combination, both tastes = 2, one taste = 1). A second Thai speaker labels a stratified 60-pair sample for kappa.

6. **Scoring.** nDCG@10 (gains 0/1/2) and P@10 per system, bootstrap CIs, paired randomization test, and breakdown by query type.

CHAPTER 5

PROGRESS AND PRELIMINARY RESULTS

5.1 Work Completed Since the Proposal

Table 5.1: Completed work since the proposal (17 September 2026).

Work item	**Result**	**Addresses**
Fair comparison	BM25 vs dense on the same unit and aggregations, with CIs and significance tests	F2a, RQ1
Volume control	Every restaurant capped at 400 reviews	F2b, RQ1
Saved evidence	Rankings and top-10 evidence stored for all runs	F3, RQ2
Thai queries	24 Thai taste queries, fixed by a native speaker before any results	F1, RQ3
Thai retrieval	Sentence pipeline (BM25 + dense), segmentation check, qualitative comparison	F1, RQ3
Thai evaluation set-up	10,000-review corpus, blind pooling and scoring procedure	F1, RQ3
Guidelines	Annotation guidelines for Thai relevance and English explanation support	F1, F3
Literature	Related work updated (Chapter 3)	F4

5.2 Fair Comparison at Review Level (RQ1)

Table 5.2 and Figure 5.1 give every model on the shared review unit.

Table 5.2: RIRD, review unit, 100 queries. nDCG@10 with 95% bootstrap CI.

Model	nDCG@10 [95% CI]	MAP	MRR	P@10	R@10
Dense, top-10 mean	**0.668** [0.625, 0.711]	**0.608**	0.760	**0.473**	**0.520**
Dense, top-5 mean	0.668 [0.625, 0.709]	0.606	0.747	0.472	0.518
Dense, top-3 mean	0.665 [0.623, 0.705]	0.601	0.744	0.470	0.517
Dense, mean	0.642 [0.598, 0.687]	0.584	0.750	0.448	0.473
Dense, max	0.625 [0.585, 0.666]	0.564	0.715	0.438	0.480
BM25, top-10 mean	0.634 [0.590, 0.676]	0.576	0.762	0.427	0.476
BM25, top-5 mean	0.621 [0.579, 0.662]	0.566	0.760	0.422	0.473
BM25, top-3 mean	0.611 [0.568, 0.654]	0.555	0.749	0.420	0.461
BM25, max	0.575 [0.532, 0.618]	0.519	0.691	0.404	0.433
BM25, mean	0.549 [0.503, 0.594]	0.507	0.636	0.377	0.423
Dense, pooled vector (proposal)	0.645 [0.602, 0.687]	0.585	0.759	0.449	0.480
BM25, concatenated doc (proposal)	0.621 [0.578, 0.663]	0.561	**0.769**	0.407	0.450
Popularity (leave-one-query-out)	0.484 [0.426, 0.540]	0.426	0.436	0.410	0.341
Review count	0.374 [0.324, 0.424]	0.349	0.505	0.281	0.202
Random (expected)	0.348 [0.311, 0.387]	0.322	0.428	0.270	0.203

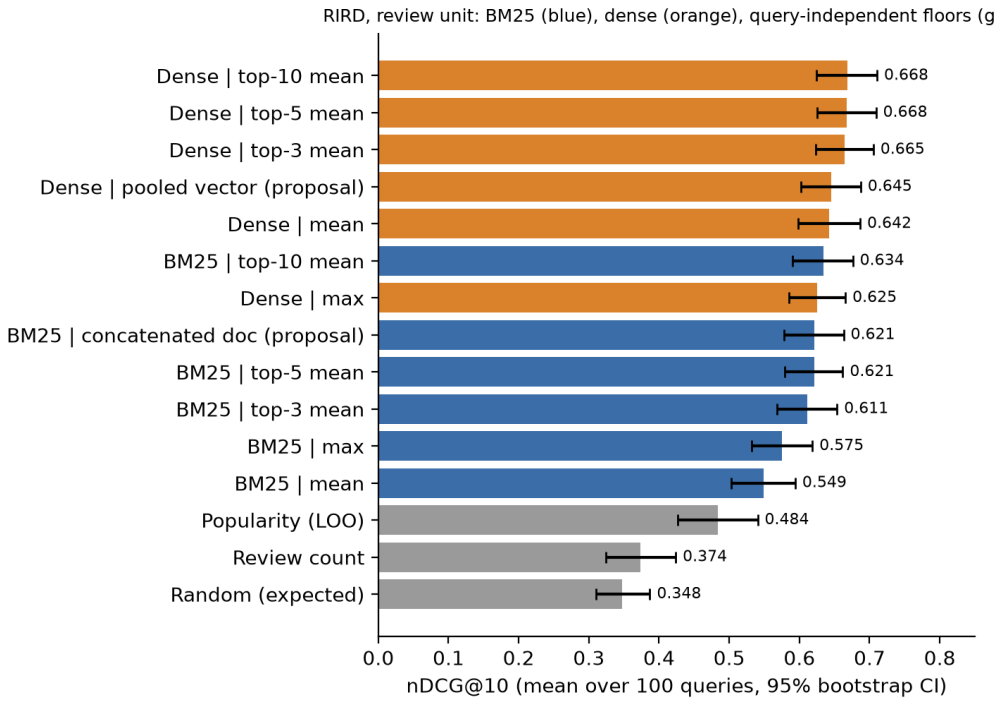


Figure 5.1: nDCG@10 with 95% bootstrap CI for all review-level models and the three query-independent floors.

Table 5.3: Dense minus BM25 on the same unit and aggregation (nDCG@10), paired randomization test, Holm-corrected over the five comparisons.

Aggregation	**Difference [95% CI]**	p **(Holm)**	**Significant**
max	+0.050 [0.017, 0.083]	0.009	yes
mean	+0.094 [0.057, 0.129]	<0.001	yes
top-3 mean	+0.053 [0.023, 0.083]	0.003	yes
top-5 mean	+0.047 [0.016, 0.077]	0.009	yes
top-10 mean	+0.035 [0.005, 0.063]	0.020	yes

Findings and interpretation.

1. **Dense beats BM25 under every aggregation**, all significant after correction (Table 5.3). The gap is largest under mean (+0.094) and smallest under top-10 mean (+0.035): averaging over all reviews hurts BM25 most because most reviews share no words with the query and score zero, whereas cosine similarity is never exactly zero.

2. **Top-k mean is the best aggregation for both models**; max is the weakest for dense. One review is a noisy signal (a single enthusiastic or off-topic review can win); averaging the best few reviews keeps the focus on query-relevant evidence while smoothing that noise.

3. **Everything except BM25 mean beats the popularity floor** (BM25 mean: +0.065, Holm $p = 0.056$). Popularity alone already reaches nDCG@10 0.484, so a large part of "accuracy" on RIRD is knowing which restaurants are relevant to many queries.

4. **Negative result: the fair setup did not significantly improve on the proposal baselines.** Dense top-10 mean is +0.023 over the proposal's pooled vector, but not significant (Holm $p = 0.64$), and BM25 mean is significantly *worse* than the proposal's concatenated document (−0.072). The point of the revision is a fair comparison, not a higher number; the proposal's conclusion (dense > BM25) survives the fix.

5. **Context.** Off-the-shelf dense with top-10 mean reaches MAP 0.608, close to the 0.626 of the fine-tuned model in Abdollah Pour et al. (2023). These are different encoders and training regimes, so this is not a like-for-like comparison, but it shows that a multilingual encoder used without training is a reasonable base for both languages.

5.3 Review-Volume Control

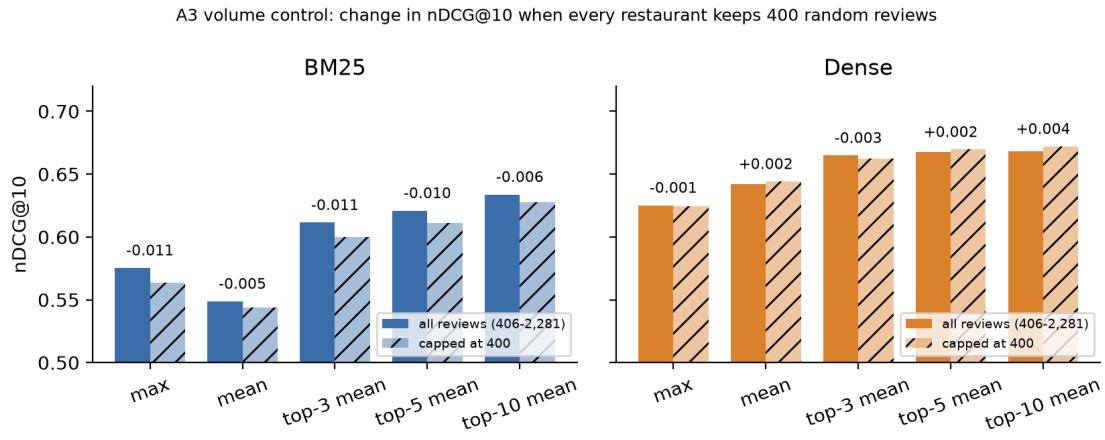


Figure 5.2: nDCG@10 with all reviews versus 400 randomly sampled reviews per restaurant. Labels give the change.

Table 5.4: Correlation between restaurant score and review count, averaged over the 100 queries (Pearson / Spearman). "Capped" correlates the capped scores with the original counts.

	BM25		Dense	
Aggregation	**all reviews**	**capped**	**all reviews**	**capped**
max	0.21 / 0.18	0.01 / 0.00	0.22 / 0.11	0.03 / −0.04
mean	−0.05 / −0.06	−0.07 / −0.07	0.04 / −0.06	0.03 / −0.06
top-3 mean	0.28 / 0.23	0.00 / −0.01	0.26 / 0.14	0.04 / −0.04
top-5 mean	0.31 / 0.26	0.00 / −0.01	0.28 / 0.15	0.04 / −0.05
top-10 mean	0.36 / 0.30	−0.01 / −0.02	0.30 / 0.16	0.04 / −0.05

Findings and interpretation.

1. **The predicted bias exists.** Within a query, max and top-k scores rise with review count (Pearson 0.21–0.36), while the mean shows none, exactly as the "more chances" argument in Section 1.3 predicts. Capping removes it.

2. **It does not change accuracy or conclusions.** Dense moves by at most ±0.004 nDCG@10; BM25 loses 0.005–0.011; dense stays best under every aggregation.

3. **Part of the bias is signal, not noise.** Restaurants with more reviews are relevant to more queries (Spearman 0.30 between review count and number of relevant queries over the 50 restaurants). BM25's small loss when capped is consistent with removing some genuine evidence: BM25 needs exact word matches, and fewer reviews means fewer matches.

4. **A lesson about measurement.** Pooling all queries into one correlation gave only 0.05–0.12, because score scales differ between queries; the per-query average shows the real effect. We report the per-query version.

5. **Limits.** One sampling seed; no paired test between the full and capped runs yet; RIRD only contains restaurants with at least 400 reviews, so smaller restaurants (where volume effects would be stronger) are not represented.

5.4 First Look at the Explanations

The two explanation tests have not been run yet, but the saved evidence lets us inspect what the system currently shows. Table 5.5 gives the top evidence for two queries under the best model.

Table 5.5: Top evidence review for the two highest-ranked restaurants (dense, top-10 mean). "Relevant" is the RIRD label.

Query	Rank: restaurant	Relevant	Top evidence (excerpt)
I am on a budget	1: Fishman Lobster Clubhouse	no	"It's not bad but kinda pricey."
	2: Dragon Boat Fusion	yes	"Good dim sum, pretty standard prices/service."
Non-spicy Chinese meals	1: Ding Tai Fung	yes	94-word review on Northern Chinese food (…) "they use a lot of oil" 2nd evidence: "…Wonton was spicy … red chili oil"
	2: Pai Northern Thai Kitchen	no	"Perfect Thai Curry in nice easy going atmosphere …"

What this shows.

- **Topic match without support.** The dense model matches the *topic* of the query (price, spiciness, Asian food) but not its *polarity*: a review saying a place is pricey ranks a restaurant first for "I am on a budget"; a sentence saying the wonton was spicy is shown for "non-spicy". These are exactly the cases the professor warned about: relevant-looking, faithful to the model, but not supporting.
- **Whole reviews are too coarse to be explanations.** A 94-word review mixes several aspects, and the part that supports the query cannot be told apart from the rest. This motivates the sentence-level rerun before the human rating.
- **Prediction.** We expect a clear share of shown evidence to be rated 0 for negation and price queries; the support rating will measure how large it is.

5.5 Thai Sentence Retrieval (RQ3, Qualitative)

Segmentation spot-check (2,000-review pilot, 19,294 sentences, 9.6 per review; checked by a native speaker). Short, informal reviews (most of Wongnai) split cleanly. Two error patterns: cuts after abbreviation dots such as กทม. and ม.ค. (0.7% of sentences) and long blog-style sections merged into one sentence (1.6% over 200 characters). Taste statements such as เผ็ดมาก or ไม่หวานมาก are short and almost always survive intact. We judge segmentation good enough for retrieval and report `crfcut`'s published accuracy on review text as a limitation (Lowphansirikul et al., 2020).

Table 5.6: Top-3 Thai sentences for one query of each type (2,000-review pilot, inspected but not yet labelled).

Query	BM25	Dense
T02 อยากกินอะไรเผ็ดๆ แบบจัดๆ (really spicy)	เนื่องจากอยากกินปูเน้นๆเนื้อๆ เลยจัดเต็ม อยากกินอะไรเบาๆและชิมชานมเย็น ถ้าอยากกินไอติมแบบอร่อย นั่งคุยกันสบายๆ	ถ้าชอบเผ็ดสั่งเพิ่มได้เลย สำหรับคนชอบเผ็ดๆเลย ต้องกินกับซอสที่ออกเปรี้ยวๆ แซลมอนรมควัน
T11 ต้มยำเปรี้ยวๆ เผ็ดๆ (sour and spicy tom yum)	น้ำต้มยำอร่อยๆ เปรี้ยวๆ ดี อร่อยดี เปรี้ยวๆ เผ็ดๆ แต่ไม่เข็ดฟันเด้อ - พิซซ่าญี่ปุ่น เปรี้ยวอย่างเดียวเปรี้ยวเผ็ดเกิน	เผ็ดเปรี้ยวนำ - ลาบหมู น้ำต้มยำเปรี้ยวแซ่บ หอมกลิ่นพริกขี้หนู ยำหมูยอแซ่บซี้ด
T20 ของหวานที่ไม่หวานมาก (desserts not too sweet)	ตอบโจทย์วัยรุ่นที่อยากกินของหวานแต่ไม่ชอบแบบหวานเลี่ยน ไม่หวานมากอย่างที่คิดไว้ ไม่หวานมากกำลังดี	และมีความหวานน้อย ขนมถ้วย : ไม่หวานมาก ไม่หวานมากแล้วก็ไม่ฝาดมาก ปล.

Observations (to be confirmed by labels).

- **BM25 is fooled by filler words.** For T02 it matches อยากกินอะไร ("want to eat something") and จัด, not spiciness. Dense finds spicy sentences.
- **Dense drifts by topic.** For T11 it returns sour-spicy sentences about other dishes (larb, yam); for T02 its fifth result describes the opposite taste (รสชาติจะออกอ่อนๆ, "mild").
- **Both handle the T20 negation**, unlike English findings on negation (Weller

et al., 2024). A possible reason is that BM25 keeps ไม่ as a token and the phrase ไม่หวานมาก is common in Thai reviews; this is a hypothesis for the labelled results.

- **The two systems are complementary.** Their top-10 lists share fewer than one sentence per query on average, so pooling them roughly doubles what must be labelled (465 pairs in the pilot) but makes the comparison informative.

5.6 Challenges

- **Review truncation.** 19.9% of RIRD reviews exceed the encoder's 256-token limit (50.6% exceeded the 128 tokens used in the proposal), so part of each long review is never seen by the dense model. This is a further reason for the sentence unit.
- **Thai labels need native speakers.** Relevance in Thai can only be judged by a native speaker, and a second independent annotator is needed for the agreement check.
- **Small Thai test set.** With 24 queries, only nDCG@10 differences of roughly 0.11–0.17 can be detected reliably; a non-significant result will be reported as "no evidence of a difference", not as "equal".

CHAPTER 6

REMAINING WORK AND PLAN

6.1 Remaining Tasks

Table 6.1: Remaining work and target weeks after the progress presentation (week 1 starts 21 October 2026).

Work item	Description	Week
Sentence-level rerun	Repeat the full RIRD comparison with sentences as the unit; repeat the volume control with more seeds and a paired test	1
Thai pool	Run retrieval on the 10,000-review corpus and build the blind labelling sheet	1
Guidelines	Final native-speaker review of the annotation guidelines	1
Deletion test	Faithfulness at sentence level, $m \in \{1, 3, 5, 10\}$, versus random deletion	2
Thai labelling	Label the pooled sentences (about 450 pairs)	2
Support rating	270 explanation items rated independently by two annotators; kappa	3
Thai agreement	Second native speaker labels 60 pairs; kappa	3
Thai scores	nDCG@10 and P@10 for BM25 vs dense, by query type	3
Final write-up	Final claims, report, slides and video	4

6.2 Expected Outcome and Fallbacks

- **RQ1** will be answered at both review and sentence level, with the volume control. We expect dense to remain ahead; whether the sentence unit helps is open.
- **RQ2** will give two numbers per model: the rank drop caused by deleting the shown evidence (versus random) and the share of shown sentences that people

rate as supporting, with kappa. A low support rate would be a valid, reportable negative finding.

- **RQ3** will give Thai nDCG@10 for BM25 and dense with CIs and a negation slice. Based on MIRACL we do not presume a winner.
- **If time runs short** we reduce the Thai test set to 15 queries, then the support rating to 15 queries. The sentence-level rerun and the deletion test are not cut, because without them the final question cannot be answered.

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